

Escaping Collapse: The Strength of Weak Data for Large Language Model Training*

Kareem Amin¹ Sara Babakniya² Alex Bie¹
Weiwei Kong¹ Umar Syed¹ Sergei Vassilvitskii¹

¹Google Research, New York

²University of Southern California, Los Angeles

Abstract

Synthetically-generated data plays an increasingly larger role in training large language models. However, while synthetic data has been found to be useful, studies have also shown that without proper curation it can cause LLM performance to plateau, or even “collapse”, after many training iterations. In this paper, we formalize this question and develop a theoretical framework to investigate how much curation is needed in order to ensure that LLM performance continually improves. Our analysis is inspired by boosting, a classic machine learning technique that leverages a very weak learning algorithm to produce an arbitrarily good classifier. The approach we analyze subsumes many recently proposed methods for training LLMs on synthetic data, and thus our analysis sheds light on why they are successful, and also suggests opportunities for future improvement. We present experiments that validate our theory, and show that dynamically focusing labeling resources on the most challenging examples — in much the same way that boosting focuses the efforts of the weak learner — leads to improved performance.

1 Introduction

Large Language Models (LLMs) represent the frontier of artificial intelligence, and are trained on vast amounts of human-generated data. However, much of the high-quality publicly available data on the Internet has been exhausted, and limits on generating new tokens threaten to slow progress on LLM training.

As a consequence, synthetically-generated datasets are playing an important role in the training of LLMs. Synthetic data have been shown to improve the performance of real large models on a range of tasks [Bai et al., 2022, Zelikman et al., 2022, Gulcehre et al., 2023, Singh et al., 2024]. On the other hand, the circuitous nature of training new LLMs on data generated by previous generations of LLMs has caused concerns of model collapse [Shumailov et al., 2024, Alemohammad et al., 2024].

What makes synthetic data beneficial or harmful? The answer depends on the precise elements of the synthetic data recipe, and one of our main contributions is a theoretical framework that unifies existing elements of synthetic data approaches, facilitating reasoning about when they might succeed or fail.

*Authors ordered alphabetically. Author contributions are listed at the end.

Basic learning theory and empirical studies suggest that a necessary condition for avoiding model collapse is that synthetic data is curated in some way to inject signal that is exogenous to the system that produced the original data. This can come in many forms: identification of high-quality subsets of synthetic data, human rewrites of poor responses, a separate model rating the responses, *etc.* A key question is how much curation is *sufficient* to not only avoid collapse, but also to converge to an optimal LLM? Our answer, which we will make precise, is the minimum amount.

Specifically, we analyze a simple procedure for improving an LLM, in which we iteratively (1) generate synthetic responses from the model; (2) obtain additional responses from an exogenous source; and (3) train the next generation of the model with both types of responses. This procedure captures previous successful approaches for training LLMs on synthetic data [Zelikman et al., 2022, Gulcehre et al., 2023, Singh et al., 2024], and so our analysis provides an explanation for why they work. More broadly, it models the ad hoc processes employed by model developers. We show that if at least a $\beta > 0$ fraction of the non-synthetic responses (*i.e.*, the ones produced by an external signal) are correct, then the iterative procedure converges to an optimal LLM (*i.e.*, one that returns a correct response to each prompt). See Theorem 5 for the precise statement and exact convergence rate.

Connection to Boosting. At a high level, our analysis shows how to use synthetic data to focus curation on regions of the prompt space where the models perform poorly. In this way, this approach resembles AdaBoost, a classic machine learning algorithm that iteratively focuses a weak learning algorithm on training examples where previous weak hypotheses performed poorly. Unlike boosting, however, our assumptions on the data and the learning method are inverted. Instead of a weak learner, we assume access to powerful LLMs that can perfectly model an input distribution, which we call *strong learners*. However, we also assume access to only weak information about the distribution we wish to model (specifically, that $\beta > 0$), *i.e.* *weak data*. This is in contrast to traditional boosting where the algorithm has access to *strong data*, *i.e.*, independent and identically distributed (i.i.d.) examples from some target distribution.

This connection between the theory of boosting and learning from synthetic data has been largely unexamined in the existing literature. Our analysis also suggests practical ways to improve current algorithms for learning from synthetic data. In our experiments, we show that scarce curation resources are better utilized by focusing their efforts on producing responses to the most challenging prompts in the training set.

2 Related work

Training models on human-generated data only has limitations such as scalability, biases, errors, and potential privacy considerations [Kurakin et al., 2023, Singh et al., 2024, Gilardi et al., 2023, Long et al., 2024]. Longpre et al. [2024] highlights a challenge: as LLMs scale, the demand for high-quality data increases, yet access to such data becomes more restricted due to copyright and privacy constraints. Given these challenges, integrating synthetic data into training pipelines is essential but comes with risks.

Model Collapse. Several studies highlight a critical concern regarding the use of synthetic data in training LLMs, known as model collapse. This phenomenon is caused by *improper* use of synthetic data in training the model, which can cause performance degradation or even complete failure of

the model [Schaeffer et al., 2025]. Alemohammad et al. [2024], Shumailov et al. [2024], Hataya et al. [2023], Gerstgrasser et al. [2024] have empirically studied model collapse in various settings, demonstrating the detrimental effects of iterative training on only synthetic data and highlighting how this process can severely degrade model performance.

Dohmatob et al. [2024a,b], Bertrand et al. [2024], Dey and Donoho [2024], Seddik et al. [2024] study model collapse theoretically. Their results show that recursively retraining only on synthetic data causes performance degradation in different models. However, combining synthetic and labeled training data [Bertrand et al., 2024, Dey and Donoho, 2024, Seddik et al., 2024, Kazdan et al., 2024, Ferbach et al., 2024] can mitigate this performance degradation. In contrast to our work, they do not demonstrate continuous improvement toward an optimal model.

Recently, Suresh et al. [2024] and Feng et al. [2024] provide theoretical explanations for model collapse under restricted models, including Gaussian mixture models and linear classifiers. Feng et al. [2024] shows that the presence of a verifier to select more desired synthetic data can improve the performance in non-recursive settings. Firdoussi et al. [2024] extends this work to include both noisy labels and features and provide theoretical and empirical results on the impacts of combining synthetic and real data. Our results do not assume a specific learning class, instead relying on a black-box strong learning assumption.

Self Improving LLMs Self-evolving or self-improving LLMs [Tao et al., 2024] is a new research direction that leverages the model itself to generate or guide the creation of high-quality data [Wang et al., 2023, Huang et al., 2023, Gillman et al., 2024], which can then be used for fine-tuning [Yuan et al., 2024, Chen et al., 2024] or RLHF [Pace et al., 2024], enabling continuous improvement with minimum or no external intervention.

STaR [Zelikman et al., 2022] presents a bootstrapping mechanism to enhance the reasoning capabilities of LLMs by iteratively asking the model to generate step-by-step “chain-of-thought” rationales for questions, filtering out incorrect answers, fine-tuning the original model on all correct rationales, and repeating the process. ReST [Gulcehre et al., 2023] proposes a combination of self-generated data and offline reinforcement learning. The method operates iteratively in two primary phases: a “Grow” phase, where for each input (context), the LLM generates multiple outputs to expand the training dataset, and an “Improve” phase, which involves ranking and filtering this augmented dataset using a learned reward model trained on human preferences. ReST^{EM} [Singh et al., 2024] is a modified version of ReST with two main differences; they do not augment the generated data with human-generated data, and in the “Improve” step instead of fine-tuning the model in the previous iteration, they fine-tune the base model. All of the above methods can be modeled in our framework, and thus we provide a better theoretical understanding about why and when such methods can work.

Recent works [Song et al., 2025, Yang et al., 2025, Das et al., 2024] take a more theoretical approach to understand self-improving algorithms. Song et al. [2025] introduces a new metric to analyze how different components contribute to self-improvement formally. In parallel, Yang et al. [2025] studies how to optimally allocate a fixed computational budget across iterations of synthetic data generation and fine-tuning, showing that exponential growth policies outperform constant or linear ones in both theory and practice.